

Disproof of conjecture 00000009682: two algebraically independent U-numbers exist

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: Mahler's classification: transcendental numbers by approximation type into classes A, S, T, U; the measure and cardinality structure of the classes. Conjecture: Lebesgue-almost every transcendental number is an S number, and the U-number set has measure zero but continuum cardinality; the dimension of algebraically independent subsets of U numbers is exactly 1, with no two-dimensional U families. (Mahler class measure structure)

The Chinese text states the same; its last clause says literally that the algebraically independent subsets of the U-numbers have dimension exactly 1 and that there is no two-dimensional U family.

Reading. The conjecture is a conjunction of (C1) almost every transcendental number is an S-number; (C2) the set of U-numbers is Lebesgue-null and has cardinality $\mathfrak{c}$; and (C3) "the dimension of algebraically independent subsets of U numbers is exactly 1, with no two-dimensional U families". We read (C3) as the conjunction of

(C3a) the supremum of $|s|$ over the subsets s of the U-numbers that are algebraically independent over $\mathbb{Q}$ equals 1 (dimension = transcendence degree), and

(C3b) there is no two-element subset of the U-numbers that is algebraically independent over $\mathbb{Q}$.

We refute (C3a) and (C3b) separately, for real numbers and for complex numbers (Lebesgue measure on $\mathbb{R}$, resp. $\mathbb{C}$). This refutes the conjunction in both settings. We do not use (C1) or (C2).

Readings not covered. (i) A reading of "dimension" as Hausdorff dimension of a set is not treated; the text attaches "dimension" to algebraically independent subsets, so we use the algebraic (transcendence-degree) reading. (ii) Our two witnesses are U-numbers of degree 1 (both have $\omega(\cdot, 1) = \infty$). A reading restricted to U-numbers of one fixed degree $n \geq 2$ is not refuted here. (iii) The class "A" in the Definition line plays no role; we only use the U- and S-classes.

2 Definitions (Wikipedia, "Transcendental number theory", section "Mahler's classification")

Source: https://en.wikipedia.org/wiki/Transcendental_number_theory. Quoted: "Next consider the values of polynomials at a complex number x , when these polynomials have integer coefficients, degree at most n , and height at most H , with n, H being positive integers." $m(x, n, H)$ is "the minimum non-zero absolute value such polynomials take at" x ; $\omega(x, n, H) = -\log m(x, n, H)/(n \log H)$ and $\omega(x, n) = \limsup_{H \rightarrow \infty} \omega(x, n, H)$. "Suppose this is

infinite for some minimum positive integer n . A complex number x in this case is called a U number of degree n .” “If the $\omega(x, n)$ are bounded, then $\omega(x)$ is finite, and x is called an S number.” The same article: “Kurt Mahler in 1932 partitioned the transcendental numbers into 3 classes, called S, T, and U.”

Definition 1 (Lean: `values`, `mahlerM`, `mahlerOmegaH`, `mahlerOmega`). For $x \in \mathbb{C}$ and $n, H \in \mathbb{N}$, let $V(x, n, H)$ be the set of non-zero values $|P(x)|$ with $P \in \mathbb{Z}[X]$, $\deg P \leq n$ and height $\leq H$, i.e. $|a_i| \leq H$ for every coefficient a_i of P . Put $m(x, n, H) = \inf V(x, n, H)$, $\omega(x, n, H) = -\log m(x, n, H)/(n \log H)$, and $\omega(x, n) = \limsup_{H \rightarrow \infty} \omega(x, n, H)$ in the extended reals $[-\infty, \infty]$, where H runs through $\mathbb{N}$.

Definition 2 (Lean: `IsUNumber`, `IsSNumber`). $x \in \mathbb{C}$ is a *U-number* if x is transcendental over $\mathbb{Q}$ and $\omega(x, n) = +\infty$ for some integer $n \geq 1$. It is an *S-number* if it is transcendental and there is a real C with $\omega(x, n) \leq C$ for all $n \geq 1$. A real number is a U-number (S-number) when it is one as a complex number; the polynomial values are the same.

Conventions. $\deg P \leq n$ is `P.natDegree <= n`. Mathlib’s $\mathbb{N}$ contains 0, so $n \geq 1$ is required explicitly. In Lean, $\log 0 = 0$, $\log 1 = 0$ and $t/0 = 0$; these conventions affect $\omega(x, n, H)$ only for $H \leq 1$, which the $\limsup$ ignores. Since $V(x, n, H)$ is finite (Lemma 3), $\inf$ is the minimum whenever $V \neq \emptyset$. *Algebraic independence* over $\mathbb{Q}$ is Mathlib’s `AlgebraicIndependent`: the evaluation map from the multivariate polynomial ring is injective. The same Wikipedia article: “A set of numbers $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$ is called algebraically independent over a field K if there is no non-zero polynomial P in n variables with coefficients in K such that $P(\alpha_1, \alpha_2, \dots, \alpha_n) = 0$.” A *Liouville number* is Mathlib’s `Liouville x`: for every n there are integers $a, b > 1$ with $0 < |x - a/b| < b^{-n}$.

The formal conjecture (Lean: `ConjectureComplex`; `ConjectureReal` is the same over $\mathbb{R}$): (C1) $\forall^{a.e.} x$, transcendental $\Rightarrow$ S-number; (C2) the U-set has volume 0 and $\#\{x : \text{U-number}\} = \mathfrak{c}$; (C3a) $\sup\{|s| : s \subseteq U, s \text{ alg. indep. over } \mathbb{Q}\} = 1$ (`Set.encard`, in $\mathbb{N} \cup \{\infty\}$); and (C3b) no such s has $|s| = 2$.

3 Proof

Lemma 3 (`finite_polys`, `values_finite`, `mahlerM_spec`). *The integer polynomials of degree $\leq n$ and height $\leq H$ form a finite set. Hence if such a P has $P(x) \neq 0$, then $0 < m(x, n, H) \leq |P(x)|$.*

Proof. $P \mapsto (a_0, \dots, a_n)$ is injective on this set, with values in the finite box $[-H, H]^{n+1}$. So $V(x, n, H)$ is finite and non-empty, and its infimum is one of its elements, which are positive. The infimum is at most $|P(x)| \in V(x, n, H)$. $\square$

Theorem 4 (`liouville_mahlerOmega_one`, `liouville_isUNumber`). *If $x \in \mathbb{R}$ is a Liouville number, then $\omega(x, 1) = +\infty$, and x is a U-number.*

Proof. Fix $w \in \mathbb{R}$ and N . Let $k = \lceil 2w \rceil + 1$, so $k - 1 \geq 2w$ and $k - 1 \geq 0$. By Mathlib’s `Liouville.frequently_exists_num`, there are $b \geq \max(N, 2, \lceil |x| \rceil + 1)$ and $a \in \mathbb{Z}$ with $0 < |x - a/b| < b^{-k}$. Let $P = bX - a$ and $H = \max(b, |a|) \geq N$. Then $\deg P \leq 1$, P has height $\leq H$, and $0 < |P(x)| = b|x - a/b| < b^{1-k}$. By Lemma 3, $0 < m(x, 1, H) \leq |P(x)|$, so $-\log m(x, 1, H) > (k - 1) \log b$. Also $|a| \leq b|x| + |bx - a| \leq b(|x| + 1)$, so $H \leq b(|x| + 1) \leq b^2$ (as $|x| + 1 \leq b$). Hence $0 < \log H \leq 2 \log b$, and

$$\omega(x, 1, H) = \frac{-\log m(x, 1, H)}{\log H} \geq \frac{(k - 1) \log b}{2 \log b} \geq w$$

(the case $w \leq 0$ is immediate). So $\omega(x, 1, H) \geq w$ for arbitrarily large H , and $\omega(x, 1) \geq w$ for every real w ; thus $\omega(x, 1) = +\infty$. Liouville numbers are transcendental (Mathlib `Liouville.transcendental`, over $\mathbb{Z}$, hence over $\mathbb{Q}$), so x is a U-number. This matches the source: “Clearly the Liouville numbers are a subset of the U numbers.” $\square$

Lemma 5 (`eventually_transcendental_adjoin`). *Let A be $\mathbb{R}$ or $\mathbb{C}$ (in Lean: any field extension of $\mathbb{R}$) and $x \in \mathbb{R}$. Then residually many $y \in \mathbb{R}$ (all y outside a meagre set) are transcendental over the $\mathbb{Q}$ -algebra $K = \mathbb{Q}[x] \subseteq A$.*

Proof. K is countable, since $\#\mathbb{Q}[x] \leq \#\mathbb{Q} + \#\{x\} + \aleph_0$ (`Algebra.cardinalMk_adjoin_le`). The elements of A algebraic over a countable domain form a countable set (`Algebraic.countable`). Its preimage in $\mathbb{R}$ is countable, and the complement of each point is open and dense, so the complement of the preimage is residual (countable intersection). $\square$

Lemma 6 (`indep_pair`). *If $x \in \mathbb{R}$ is transcendental over $\mathbb{Q}$ and y is transcendental over $\mathbb{Q}[x]$ (in A), then $x \neq y$ and $\{x, y\} \subseteq A$ is algebraically independent over $\mathbb{Q}$.*

Proof. $\{x\}$ is independent because x is transcendental (`algebraicIndependent_unique_type_iff`). Adding an element transcendental over the algebra generated by the set keeps independence (`AlgebraicIndepOn.insert`). And $y \neq x$ since $x \in \mathbb{Q}[x]$ is algebraic over it. $\square$

Theorem 7 (`exists_two_indep_U`). *Let $x = \sum_{i \geq 0} 10^{-i!}$ (Mathlib `liouvilleNumber 10`; the sum starts at $i = 0$, so x is $1/10$ plus Liouville’s constant $\sum_{k \geq 1} 10^{-k!}$; Liouville by `liouville_liouvilleNumber`). There is a real $y \neq x$ such that x, y are U-numbers and $\{x, y\}$ is algebraically independent over $\mathbb{Q}$, both in $\mathbb{R}$ and in $\mathbb{C}$.*

Proof. The Liouville numbers form a residual set (Mathlib `eventually_residual_liouville`). Intersect it with the two residual sets of the previous lemma ($A = \mathbb{R}$ and $A = \mathbb{C}$). The intersection is residual, hence dense by the Baire category theorem (`dense_of_mem_residual`) and non-empty; pick y in it. Then y is Liouville, hence a U-number, and the previous lemma gives independence. $\square$

Theorem 8 (`not_conjectureReal`, `not_conjectureComplex`, `two_le_iSup_real`, `two_le_iSup_complex`). *The set $s = \{x, y\}$ is a two-element algebraically independent set of U-numbers, in $\mathbb{R}$ and in $\mathbb{C}$ (`two_dim_U_family_real`, `two_dim_U_family_complex`). So (C3b) fails, the supremum in (C3a) is ≥ 2 , and *ConjectureReal* and *ConjectureComplex* are false.*

4 Lean formalization

File `lean/Conjecture9682/Basic.lean` (300 lines; only import Mathlib, Lean v4.33.1 with Mathlib v4.33.1). Main theorems: (Unicode symbols transliterated to ASCII.)

```
theorem not_conjectureComplex : Not ConjectureComplex
theorem not_conjectureReal : Not ConjectureReal
theorem two_le_iSup_complex : 2 <= iSup (fun s : {s : Set Complex //
  s subset {x | IsUNumber x} and
  AlgebraicIndependent Rat (Subtype.val : s -> Complex)} => s.1.encard)
theorem exists_two_indep_U : exists x y : Real, x != y and
  IsUNumber (x : Complex) and IsUNumber (y : Complex) and
  AlgebraicIndependent Rat (Subtype.val : ({x, y} : Set Real) -> Real) and
  AlgebraicIndependent Rat
  (Subtype.val : ({(x : Complex), (y : Complex)} : Set Complex) -> Complex)
```

Axiom audit (#print axioms) for not_conjectureComplex, not_conjectureReal, two_le_iSup_complex, two_le_iSup_real and liouville_isUNumber: [propext, Classical.choice, Quot.sound]. There is no sorry and no native_decide.